

Mingu Hwang

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Education

Stony Brook University

Stony Brook, New York

Bachelor of Science, Computer Science
Dean's List for Fall 2024

Expected May 2027

Professional Experience

SM Solutions

Suwon, Korea

Intern

November 2025 – January 2026

- Developed a Python pipeline using PyMuPDF to convert MISRA C PDFs into structured Markdown, also for the tables as markdown tables to prepare company documents for an internal LLM.
- Implemented regex-based hierarchical chunking by heading patterns and indexed sentence-level embeddings (mxbai-embed-large-v1) in ChromaDB, mapped back to parent chunks.
- Built a keyword-based search system that falls back to semantic vector search and used cosine-similarity threshold to reject out-of-domain queries.

MindSyn

Incheon, Korea

Intern

June 2025 – August 2025

- Ported a Spike-Driven Transformer from Python (PyTorch/SpikingJelly) to dependency-free C and implemented on a KEIT neuromorphic chip, using convolution, batch normalization, LIF spiking neurons, and spike-driven self-attention without ML libraries.
- Built a PyTorch-to-C weight export pipeline and verified C inference against Python reference outputs, matching the original model's high accuracy on MNIST.

Brain-Inspired Computing Lab – Prof. Yoon Seok Yang

Incheon, Korea

Undergraduate Research Assistant

March 2025 – August 2025

- Studied and presented about the Spike-Driven Transformer (NeurIPS 2023), analyzing its multiplication-free spike-driven self-attention and spiking patch splitting modules for energy-efficient neuromorphic computing.

Leadership Experience

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Incheon, Korea

Programming Team leader

August 2023 – June 2025

- Led a team of 7 tutors to design and teach a 7-week Java curriculum, along with introductory Python and Scratch courses, for elementary through high school students.
- Managed project-based lessons with hands-on debugging practice, guiding students to create their own projects inspired by real-life problems.

2024 Incheon Global STEAM Education

Incheon, Korea

Mentor

September 2024 – December 2024

- Taught computational thinking to elementary school students through Scratch workshops.
- Guided students to program geometry formulas, including triangle, rectangle, and circle area and circumference, connecting math concepts to real-world problem solving.

Skills

Programming: Python, C, Java, SQL

AI/ML: LLM (Sentence Transformers, PyMuPDF, mxbai-embed-large-v1)/RAG, Vector Database (ChromaDB)

Development Tools: Docker, Git, Linux

Language: Proficient in English, Native Proficiency in Korean

Certifications: ADsP (Advanced Data Analytics Semi-Professional), SQLD (SQL Developer), Linux Master Level 2, Information Processing Industrial Engineer – Korean national certifications